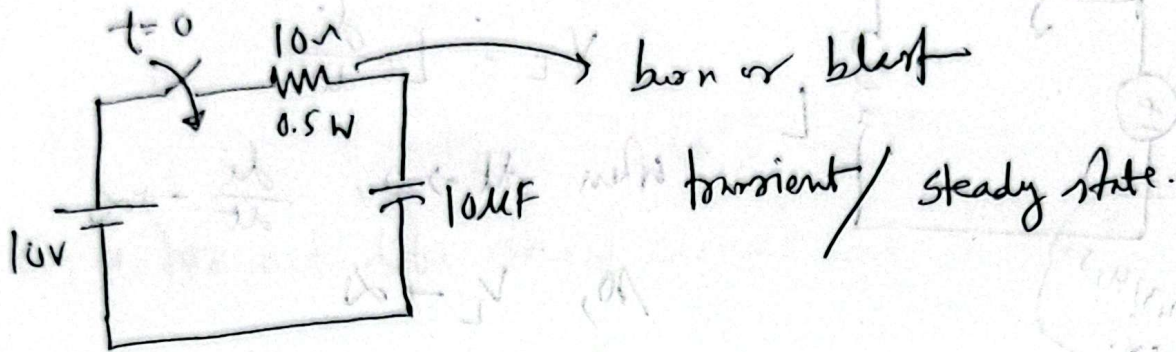


Transient Analysis

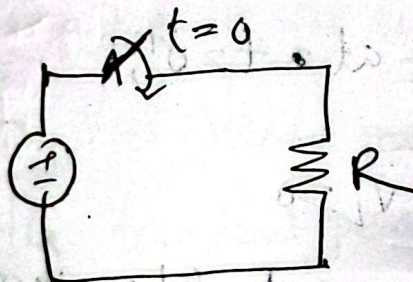


Behaviour of Basic Component

$t=0^- \rightarrow$ just before the switch is closed

$t=0 \rightarrow$ when the switch is closed

$t=0^+ \rightarrow$ just after the switch is closed.



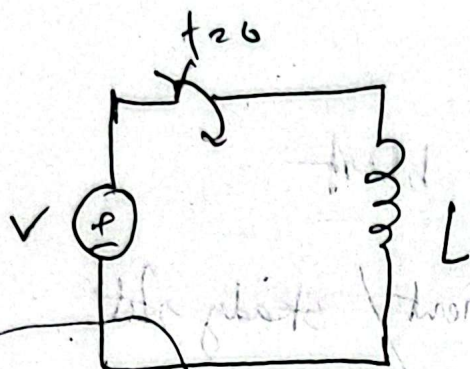
at $t=0^- \Rightarrow I=0$

at $t=0 \Rightarrow I = \frac{V}{R}$

at $t=0^+ \Rightarrow I = \frac{V}{R}$

i.e, Resistance can ~~not~~ abruptly change its voltage and current.

⑦ What is the behavior at $t = 1/4, 1/5$ we know later.



at, $t=0$,

$$V_L = L \frac{di}{dt}$$

When $dt \rightarrow 0$, $\frac{di}{dt} \rightarrow \infty$

So, $V_L \rightarrow \infty$

Which is not valid.

So, i.e., current through an inductor can not change abruptly/instantaneously.

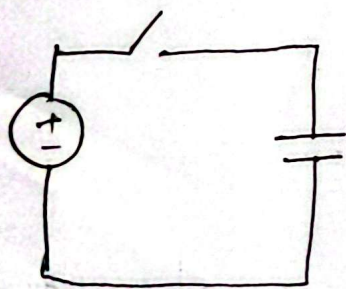
$$i_L(0^-) = i_L(0^+)$$

Now, $i_L(0^-) = 0A$, So, $i_L(0^+) = 0A$, inductor act like an open circuit at $t=0^+$.

Now, at $dt \rightarrow \infty$, $\Rightarrow V_L = 0$

at Steady state inductor act like short circuit.

————— X —————



at $t=0$,

$$i_C = C \frac{dv}{dt}$$

When

$dt \rightarrow 0$, $\frac{dv}{dt} \rightarrow \infty$

and $i_C \rightarrow \infty$ not valid.

Voltage across a capacitor cannot change instantaneously.

$$\therefore V_C(0^+) = V_C(0^-)$$

1] $V_C(0^-) = 0V$ then, $V_C(0^+) = 0V$.

Capacitor act like short circuit at $t = 0^+$.

Now, at $dt \rightarrow \infty$, $I_C = 0A$,

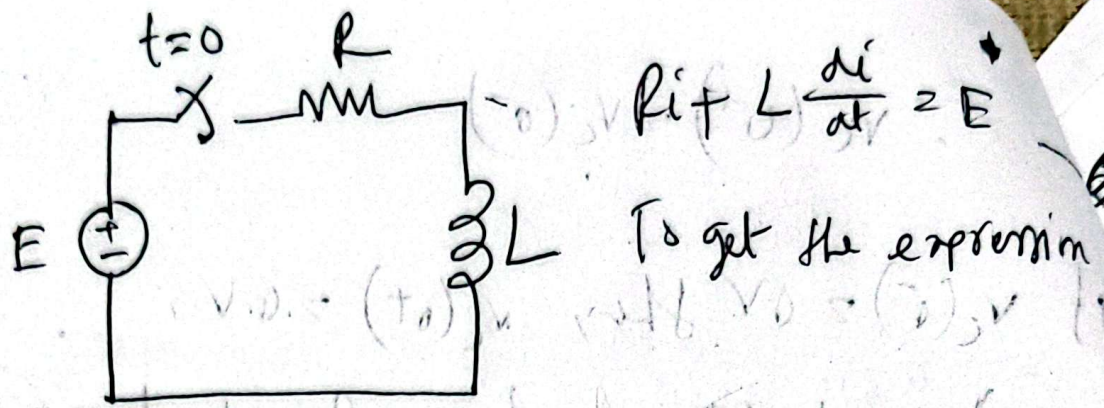
at steady state capacitor act like an open circuit.

Transient:

The period required for the current or voltage to adjust themselves to their steady state modes of variation is called the transient period.

Causes of transient:

1. Sudden load change
2. Faults in the power system.
3. Switching operations.
4. Lightning.



Way 1:

$$L \frac{di}{dt} = E - Ri$$

$$\Rightarrow \frac{L}{E - Ri} di = dt$$

$$\Rightarrow \int \frac{L}{E - Ri} di = \int dt$$

$$\Rightarrow -\frac{L}{R} \log_e (E - Ri) = t + c$$

$$\Rightarrow \log_e (E - Ri) = -\frac{R}{L} t + c_1$$

$$\Rightarrow E - Ri = A e^{-\frac{R}{L} t}$$

$$= A_1 e^{-\frac{R}{L} t}$$

$$\Rightarrow i(t) = \frac{E}{R} - A e^{-\frac{R}{L} t}$$

At $t=0, i=0, A = \frac{E}{R}$

$$i(t) = \frac{E}{R} - \frac{E}{R} e^{-\frac{R}{L}t}$$

↓
steady state

↓
transient.

Way 2:

$$Ri + L \frac{di}{dt} = E$$

Simple / capacitor (no
circuit in Q1 not Q2)

Transient (independent of the source)

$$Ri + L \frac{di}{dt} = 0$$

let, $i = Ae^{st}$

$$R Ae^{st} + L \frac{d}{dt} (Ae^{st}) = 0$$

$$\Rightarrow R Ae^{st} + ALs e^{st} = 0$$

$$\Rightarrow R + Ls = 0$$

$$\therefore s = -\frac{R}{L}$$

$$i_h = Ae^{-\frac{R}{L}t}$$

Steady State,

$$dt \rightarrow d,$$

$$\Rightarrow \frac{di}{dt} = 0$$

$$Ri_{ss} = E$$

$$\therefore i_{ss} = \frac{E}{R}$$

B.C

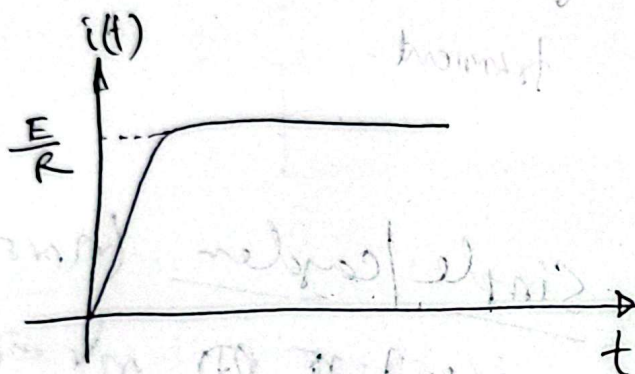
$$t = 0, i = 0$$

$$0 = \frac{E}{R} + Ae^{-0}$$

$$A = -\frac{E}{R}$$

$$i(t) = \frac{E}{R} - \frac{E}{R} e^{-\frac{R}{L}t}$$

$$i(t) = \frac{E}{R} - \frac{E}{R} e^{-R/L t}$$



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linear circuit analysis

at, $t=0$, $i(0) = \frac{E}{R} - \frac{E}{R} = 0 \text{ A}$,

at, $t \rightarrow \infty$, $i(t) = \frac{E}{R} - \frac{E}{R} \cdot \frac{1}{e^{\infty}} = \frac{E}{R}$.

at, $t = \frac{L}{R}$; $i = \frac{E}{R} (1 - e^{-1}) = 0.632 \frac{E}{R}$.

So, at $t = \frac{L}{R}$, 'i' reaches 63.2% of its final change.

$t = \frac{L}{R}$, is called the time constant of RL circuit.

$$\therefore i(t) = \frac{E}{R} (1 - e^{-t/\tau})$$

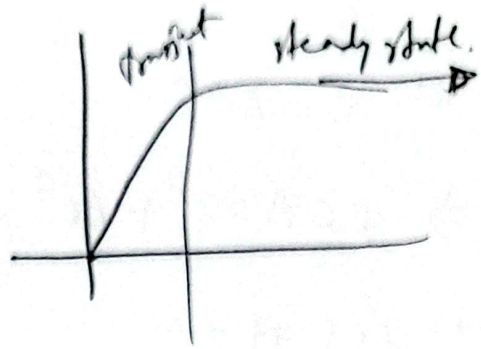
Ans.

$t = \tau$

$$i(t) = \frac{E}{R} - \frac{E}{R} e^{-\frac{R}{L}t}$$

$$= \frac{E}{R} (1 - e^{-\frac{R}{L}t})$$

$$= \frac{E}{R} (1 - e^{-\frac{t}{\tau}})$$



$$[t = 0.001 \tau; i = 9.99 \times 10^{-4}, \quad t = 0.01 \tau; i = 9.95 \times 10^{-3}]$$

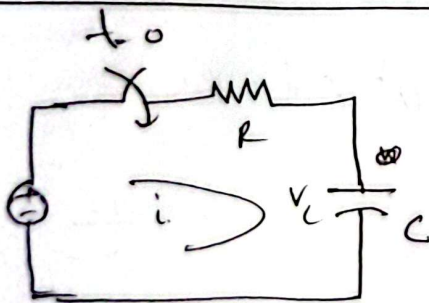
$$at, \quad t = \tau, \quad i(\tau) = 0.632 \frac{E}{R}. \quad [t = 0.1, i = 0.09]$$

$$t = 2\tau, \quad i(t) = 0.864 \frac{E}{R}.$$

$$t = 3\tau, \quad i(t) = 0.95 \frac{E}{R}$$

$$t = 4\tau, \quad i(t) = 0.98 \frac{E}{R},$$

$$t = 5\tau, \quad i(t) = 0.99 \frac{E}{R}.$$



$$Ri + \frac{1}{C} \int i dt = E$$

But to derive the expression of V_c

$$RC \frac{dV_c}{dt} + V_c = E$$

Transient

$$RC \frac{dV_c}{dt} + V_c = 0$$

Let, $V_c = A e^{st}$

$\Rightarrow RCAs e^{st} + Ae^{st} = 0$

$\Rightarrow RCs + 1 = 0$

$s = -\frac{1}{RC}$

$\therefore V_c = A e^{-\frac{t}{RC}}$

Steady state,

$dt \rightarrow \infty$

$V_c(\infty) = E$

$\therefore V_c(t) = E + A e^{-\frac{t}{RC}}$

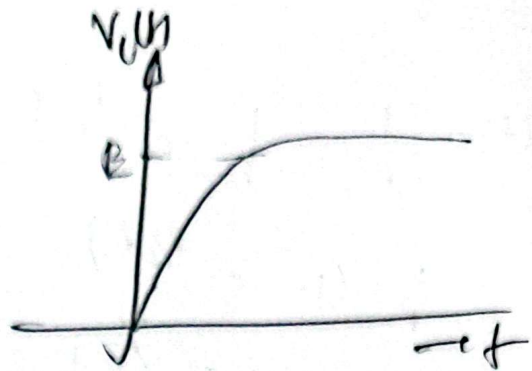
B.C at, $t = 0$

$V_c(t) = 0$

$\therefore 0 = E + A$

$\therefore A = -E$

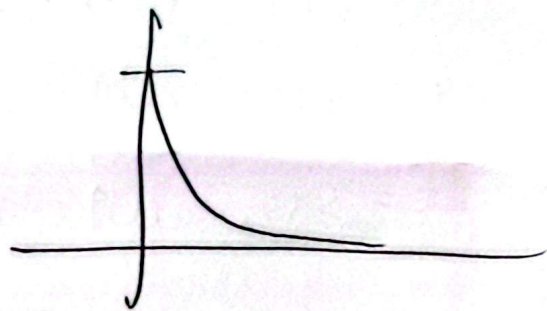
$\therefore V_c(t) = E - E e^{-\frac{t}{RC}}$
 $= E(1 - e^{-\frac{t}{RC}})$



$i_c = C \frac{dV_c}{dt}$

$= C \times \frac{1}{RC} \cdot E e^{-\frac{t}{RC}}$

$= \frac{E}{R} e^{-\frac{t}{RC}}$



at, $t = RC$

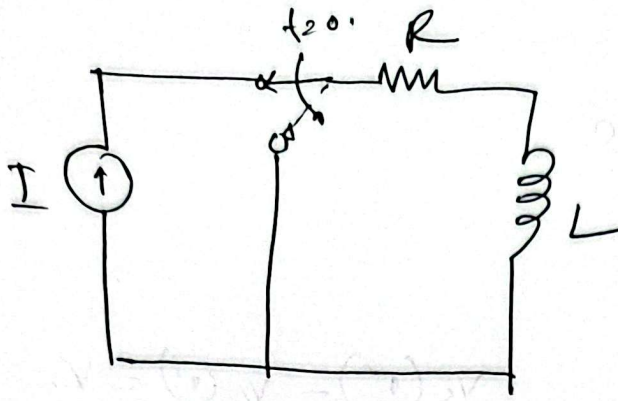
$i = 0.367 \frac{E}{R}$

Now, at, $t = RC$, $i = 36.7\%$

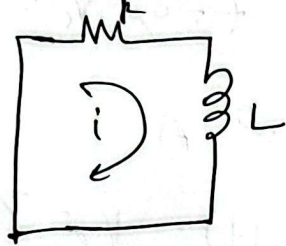
of total current,

$t = RC$ is known as the time constant of the circuit $= \tau$

$i(t) = \frac{E}{R} e^{-\frac{t}{\tau}}$



$$i_L(0^-) = I$$



$$i_L(0^-) = I$$

$$i_L(0^+) = I$$

B.C

$$\text{at, } i_L(0) = I$$

$$A = I$$

$$\therefore i(t) = I e^{-\frac{R}{L}t}$$

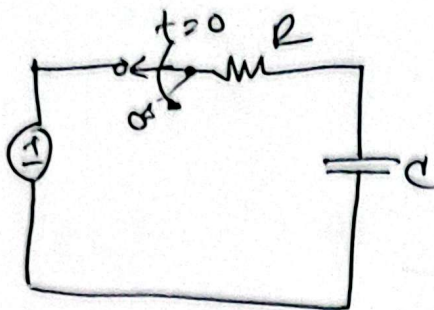
$$Ri + L \frac{di}{dt} = 0$$

$$\text{Let, } i = A e^{st}$$

$$R + Ls = 0$$

$$s = -\frac{R}{L}$$

$$\therefore i(t) = A e^{-\frac{R}{L}t}$$



$$RC \frac{dV_C}{dt} + V_C = 0$$

$$V_C(0^-) = V_C(0^+) = V$$

$$\text{let } V_C = Ae^{st}$$

$$\text{at } t=0, V_C = V$$

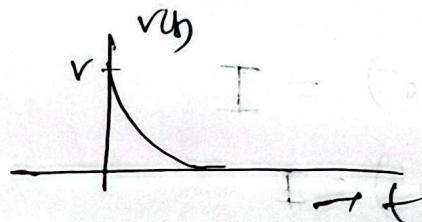
$$RCs + 1 = 0$$

$$V = Ae^{0} = A$$

$$s = -\frac{1}{RC}$$

$$V_C(t) = V e^{-t/RC}$$

$$V_C(t) = A e^{-t/RC}$$



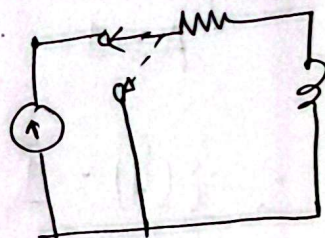
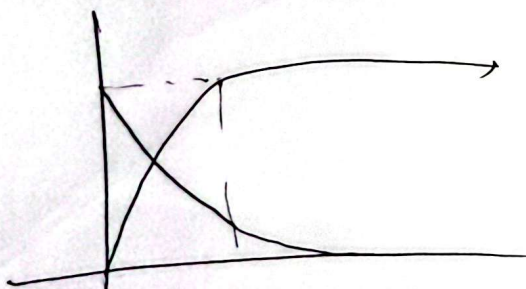
Summary d:



charging

$$i_L(t) = \frac{E}{R} (1 - e^{-\frac{R}{L}t})$$

$$V_L(t) = L \frac{di}{dt} = E e^{-\frac{R}{L}t}$$



$$i_L(t) = I e^{-\frac{R}{L}t}$$

$$V_L(t) = -IR e^{-\frac{R}{L}t} = -V_R$$

